

Investing Money - Simple Interest - Changing the Term

Simple Interest is called simple interest because at the end of each year no interest is paid, it is only paid when the total time period of the investment is reached.

$$I = Prn$$

n = the term, which is the time period the investment (or loan) is over

The term can be any period, but because the interest rate is usually an annual (yearly) rate the term should also be in years. So when you have an investment period of days, weeks or months, these have to be converted to years using fractions.

To convert to years:

<u>Number of Days</u>	<u>Number of Weeks</u>	<u>Number of Months</u>
365	52	12
Period in Days	Period in Weeks	Period in Months

Note that the denominators (bottom numbers) don't change if the period is longer than a year. For example if a period is 15 months (more than 1 year), you still use 12, not 24.

Example: Find the interest earned when \$5 100 is invested
for 17 months at 7% p.a.

$r = 7 \div 100 = 0.07$	←	Convert 7% to a decimal by $\div 100$
$I = Prn$	←	Write the formula
$= 5\,100 \times 0.07 \times \frac{17}{12}$	←	Substitute values into formula Changing 17 months into a fraction of a year.
$I = \$505.75$	←	
Answer with '\$' sign and cents to 2 decimal places (if needed)		

Investing Money - Simple Interest - Changing the Term (Sheet 1)

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Write these time periods as fractions of a year. Don't simplify the fractions.

Examples:

$$5 \text{ months} = \frac{5}{12} \quad 13 \text{ weeks} = \frac{13}{52}$$

1 25 weeks = $\frac{\quad}{\quad}$

2 9 months = $\frac{\quad}{\quad}$

3 117 days = $\frac{\quad}{\quad}$

4 14 weeks = $\frac{\quad}{\quad}$

5 15 months = $\frac{\quad}{\quad}$

Calculate the interest earned on these investments.

$P = \$1\,040$, $R = 8\%$ p.a., $n = 9$ weeks

6 $r = \frac{\quad}{\quad} \div 100 = \frac{\quad}{\quad}$

Convert the rate to a decimal

$I = Prn$

$$= \frac{\quad}{\quad} \times \frac{\quad}{\quad} \times \frac{\quad}{\quad}$$

P r n

Calc: $1040 \times 0.08 \times 9 \div 52 =$

$I =$ $\frac{\quad}{\quad}$

Show the '\$' sign and write cents to 2 d.p.

$P = \$4\,260$, $R = 9\%$ p.a., $n = 7$ months

7 $r = \frac{\quad}{\quad} \div 100 = \frac{\quad}{\quad}$

$I = Prn$

$$= \frac{\quad}{\quad} \times \frac{\quad}{\quad} \times \frac{\quad}{\quad}$$

$I =$ $\frac{\quad}{\quad}$

$P = \$1\,095$, $R = 5.6\%$ p.a., $n = 220$ days

8 $r = \frac{\quad}{\quad} \div \frac{\quad}{\quad} = \frac{\quad}{\quad}$

$I = Prn$

$$= \frac{\quad}{\quad} \times \frac{\quad}{\quad} \times \frac{\quad}{\quad}$$

$I =$ $\frac{\quad}{\quad}$

$P = \$7\,176$, $R = 3.2\%$ p.a., $n = 85$ wks.

9 $r =$ $\frac{\quad}{\quad}$

$I = Prn$

$$= \frac{\quad}{\quad} \times \frac{\quad}{\quad} \times \frac{\quad}{\quad}$$

$I =$ $\frac{\quad}{\quad}$



Complete these calculations, rounding to 2 decimal places.

10 $I = 5\,000 \times 0.046 \times \frac{19}{12}$

$I =$ $\frac{\quad}{\quad}$

11 $I = 83\,560 \times 0.105 \times \frac{96}{365}$

$I =$ $\frac{\quad}{\quad}$

$P = \$2\,500$, $R = 6.3\%$ p.a., $n = 177$ days

12 $I = Prn$

$$= \frac{\quad}{\quad} \times \frac{\quad}{\quad} \times \frac{\quad}{\quad}$$

r as a decimal

$I =$ $\frac{\quad}{\quad}$

$P = \$880$, $R = 13.25\%$ p.a., $n = 16$ mths

13 $I =$ $\frac{\quad}{\quad}$



Now find the interest and then the account balance.

$P = \$700$, $R = 5\%$ p.a., $n = 12$ weeks

14 $I =$ $\frac{\quad}{\quad}$

$A =$ $\frac{\quad}{\quad} + \frac{\quad}{\quad}$

Principal Interest

$A =$ $\frac{\quad}{\quad}$

$P = \$34\,900$, $R = 11.6\%$ p.a.,
 $n = 20$ months.

15 $I =$ $\frac{\quad}{\quad}$

$A =$ $\frac{\quad}{\quad}$

$A =$ $\frac{\quad}{\quad}$

Ben invests \$8 500 at 12.2% p.a. for 34 weeks. He wants \$10 000 for a new car. Calculate:

16 His interest and Acc. Bal.

17 The amount above or below \$10 000.

Use the words 'above' or 'below'.

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Find the interest earned and the account balance with these.

Tom invests \$17 450 at 6.65% p.a. for 10 months.

1 Convert 6.65% to a decimal.

$$r = \boxed{} \div \boxed{} = \boxed{}$$

2 Calculate the interest earned

$$I = Prn$$

$$= \boxed{} \times \boxed{} \times \boxed{}$$

$$I =$$

Show the '\$' sign and write cents to 2 d.p.

3 Find his account balance after the interest has been paid.

$$A = \boxed{} + \boxed{}$$

$$A =$$

Josiah invests \$4 200 in a term deposit at 8.35% p.a. for 195 days.

4 Convert 8.35% to a decimal.

5 Calculate the interest earned.

6 Find his account balance after the interest is credited.

Bailey is saving for a trip. He invests \$11 200 for 15 months at 7.16% p.a.

7 Express 7.16% as a decimal.

8 Calculate the interest earned.

9 Find his account balance after the interest has been paid.

10 If the airfare is \$2 120, how much spending money is left?

Julia has been left an inheritance. Her father gives her a choice, she can:
1 - Receive \$19 000 now, and invest it at 4.35% p.a. for 75 weeks, or
2 - Receive \$21 000 in 75 weeks time. Which is the best financial option?

11 Calculate the interest earned on the \$19 000 investment (1).

12 Find Julia's account balance after the interest has been paid.

13 Which option is the best? 1 or 2 and by how much?

Tayla won a prize of \$2 000 and she invested it for 175 days at 4.3% p.a.

14 Calculate the interest Tayla earned on the \$2 000 investment.

15 Find Tayla's account balance when the interest was deposited.

16 If she spends 75% of the money on a new laptop, how much did the laptop cost?

$$\text{Cost} = \boxed{} \times \boxed{}$$

$$\text{Cost} = \boxed{}$$

Wai received \$765 for holiday work and \$320 on his birthday. He invests all the money for 17 weeks at 3.3% p.a. Find the interest Wai earned.

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Write these time periods as fractions of a year. Don't simplify the fractions.

Examples:

$$5 \text{ months} = \frac{5}{12} \quad 13 \text{ weeks} = \frac{13}{52}$$

1 25 weeks = $\frac{25}{52}$

2 9 months = $\frac{9}{12}$

3 117 days = $\frac{117}{365}$

4 14 weeks = $\frac{14}{52}$

5 15 months = $\frac{15}{12}$

Calculate the interest earned on these investments.

$P = \$1\,040$, $R = 8\%$ p.a., $n = 9$ weeks

6 $r = \frac{8}{100} = 0.08$

Convert the rate to a decimal

$I = Prn$

$$= \frac{1\,040}{P} \times \frac{0.08}{r} \times \frac{9}{n}$$

Calc: $1\,040 \times 0.08 \times 9 \div 52 =$

$I = \$14.40$

Show the '\$' sign and write cents to 2 d.p.

$P = \$4\,260$, $R = 9\%$ p.a., $n = 7$ months

7 $r = \frac{9}{100} = 0.09$

$I = Prn$

$$= 4\,260 \times 0.09 \times \frac{7}{12}$$

$I = \$223.65$

$P = \$1\,095$, $R = 5.6\%$ p.a., $n = 220$ days

8 $r = \frac{5.6}{100} = 0.056$

$I = Prn$

$$= 1\,095 \times 0.056 \times \frac{220}{365}$$

$I = \$36.96$

$P = \$7\,176$, $R = 3.2\%$ p.a., $n = 85$ wks.

9 $r = \frac{3.2}{100} = 0.032$

$I = Prn$

$$= 7\,176 \times 0.032 \times \frac{85}{52}$$

$I = \$375.36$

Complete these calculations, rounding to 2 decimal places.

10 $I = 5\,000 \times 0.046 \times \frac{19}{12}$

$I = (\$)364.17$

11 $I = 83\,560 \times 0.105 \times \frac{96}{365}$

$I = (\$)2\,307.63$

$P = \$2\,500$, $R = 6.3\%$ p.a., $n = 177$ days

12 $I = Prn$

$$= 2\,500 \times 0.063 \times \frac{177}{365}$$

r as a decimal

$I = \$76.38$

$P = \$880$, $R = 13.25\%$ p.a., $n = 16$ mths

13 $I = Prn$

$$= 880 \times 0.1325 \times \frac{16}{12}$$

$I = \$155.47$



Now find the interest and then the account balance.

$P = \$700$, $R = 5\%$ p.a., $n = 12$ weeks

14 $I = Prn$

$$= 700 \times 0.05 \times \frac{12}{52}$$

$I = \$8.08$

$A = \frac{700}{\text{Principal}} + \frac{8.08}{\text{Interest}}$

$A = \$708.08$

$P = \$34\,900$, $R = 11.6\%$ p.a.,
 $n = 20$ months.

15 $I = Prn$

$$= 34\,900 \times 0.116 \times \frac{20}{12}$$

$I = \$6\,747.33$

$A = 34\,900 + 6\,747.33$

$A = \$41\,647.33$

Ben invests \$8 500 at 12.2% p.a. for 34 weeks. He wants \$10 000 for a new car. Calculate:

16 His interest and Acc. Bal.

$I = Prn$

$$= 8\,500 \times 0.122 \times \frac{34}{52}$$

$I = \$678.04$

$A = 8\,500 + 678.04$

$A = \$9\,178.04$

17 The amount above or below \$10 000.

Amount = $10\,000 - 9\,178.04$

Amount = \$821.96 below

Use the words 'above' or 'below'.

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 Find the interest earned and the account balance with these. Tom invests \$17 450 at 6.65% p.a. for 10 months. 1 Convert 6.65% to a decimal. $r = \boxed{6.65} \div \boxed{100} = \boxed{0.0665}$ 2 Calculate the interest earned $I = Prn$ $= \boxed{17\,450} \times \boxed{0.0665} \times \frac{\boxed{10}}{\boxed{12}}$ $I = \\$967.02$ Show the '\$' sign and write cents to 2 d.p. 3 Find his account balance after the interest has been paid. $A = \boxed{17\,450} + \boxed{967.02}$ $A = \\$18\,417.02$ Josiah invests \$4 200 in a term deposit at 8.35% p.a. for 195 days. 4 Convert 8.35% to a decimal. $r = 8.35 \div 100 = 0.0835$ 5 Calculate the interest earned. $I = Prn$ $= 4\,200 \times 0.0835 \times \frac{195}{365}$ $I = \\$187.36$ 6 Find his account balance after the interest is credited. $A = 4\,200 + 187.36$ $A = \\$4\,387.36$	Bailey is saving for a trip. He invests \$11 200 for 15 months at 7.16% p.a. 7 Express 7.16% as a decimal. $r = 7.16 \div 100 = 0.0716$ 8 Calculate the interest earned. $I = Prn$ $= 11\,200 \times 0.0716 \times \frac{15}{12}$ $I = \\$1\,002.40$ 9 Find his account balance after the interest has been paid. $A = 11\,200 + 1\,002.40$ $A = \\$12\,202.40$ 10 If the airfare is \$2 120, how much spending money is left? $S. Money = 12\,202.40 - 2\,120$ $S. Money = \\$10\,082.40$ Julia has been left an inheritance. Her father gives her a choice, she can: 1 - Receive \$19 000 now, and invest it at 4.35% p.a. for 75 weeks, or 2 - Receive \$21 000 in 75 weeks time. Which is the best financial option? 11 Calculate the interest earned on the \$19 000 investment (1). $I = Prn$ $= 19\,000 \times 0.0435 \times \frac{75}{52}$ $I = \\$1\,192.07$ 12 Find Julia's account balance after the interest has been paid. $A = 19\,000 + 1\,192.07$ $A = \\$20\,192.07$	13 Which option is the best? 1 or 2 and by how much? Option 2 is the best option $Amount = 21\,000 - 20\,192.07$ $Amount = \\$807.93$ Tayla won a prize of \$2 000 and she invested it for 175 days at 4.3% p.a. 14 Calculate the interest Tayla earned on the \$2 000 investment. $I = Prn$ $= 2\,000 \times 0.043 \times \frac{175}{365}$ $I = \\$41.23$ 15 Find Tayla's account balance when the interest was deposited. $A = 2\,000 + 41.23$ $A = \\$2\,041.23$ 16 If she spends 75% of the money on a new laptop, how much did the laptop cost? $Cost = \boxed{2\,041.23} \times \frac{\boxed{75}}{\boxed{100}}$ $Cost = \boxed{\\$1\,530.92}$ Wai received \$765 for holiday work and \$320 on his birthday. He invests all the money for 17 weeks at 3.3% p.a. Find the interest Wai earned. 17 Total = 765 + 320 $Total = \\$1\,085$ $I = Prn$ $= 1\,085 \times 0.033 \times \frac{17}{52}$ $I = \\$11.71$
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